MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. A compound pendulum of mass m consists of a symmetric rigid body (shown in Figure 1.1) and a horizontal massless plate (not shown in Figure 1.1). The plate is attached to point O in Figure 1.1 and is free to rotate by an angle ψ about the vertical Earth-fixed axis through O. The rigid body is fixed at one of its end points along its axis of symmetry (point O in Figure 1.1) to the plate and is free to rotate by an angle θ about a horizontal plate-fixed axis through O, under the influence of gravity. The centre of mass of the rigid body is on the axis of symmetry at distance f from point O.

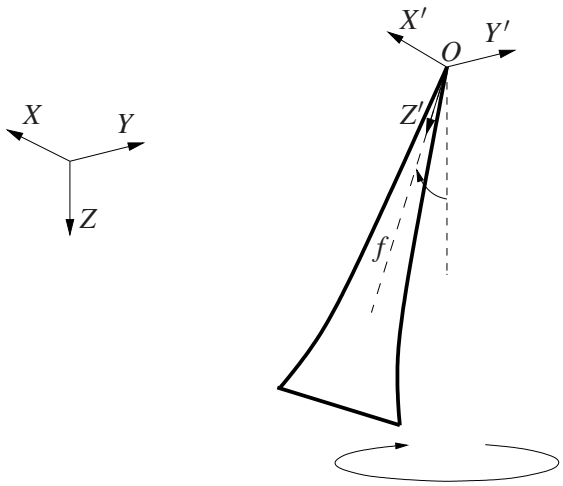


Figure 1.1 Compound 2-degree-of-freedom pendulum.

A body-fixed Cartesian coordinate system has its origin at O, and its z-axis (Z') along the axis of symmetry of the body and its y-axis (Y') along the horizontal plate-fixed axis about which the rigid body rotates by the angle θ. The body-fixed x-, y-, and z-axes ($X'Y'Z'$) correspond to principal axes with principal moments of inertia I_{xx}, I_{yy} ($= I_{xx}$), and I_{zz} respectively. In the nominal configuration the body-fixed x-, y-, and z-axes are respectively aligned with the x-, y-, and z-axes (XYZ) of an Earth-fixed Cartesian coordinate system, in which the x-axis is horizontal and the z-axis points downwards. The angular configuration of the body is therefore given by the Euler angle convention, first rotation from its nominal configuration by angle ψ followed by angle θ rotation.

a) Calculate the body angular velocity vector in the body-fixed coordinate system in terms of the generalised coordinates. [5]

b) Compute the kinetic energy of the compound pendulum. [5]

c) Calculate the potential energy of the compound pendulum, and hence determine the Lagrangian function. [4]

d) Use the Lagrangian approach to derive the equations of motion of the compound pendulum in terms of the generalised coordinates. [6]

2. It is desired to achieve lateral balancing of a humanoid robot by antisymmetric rotation of the robot arms in a vertical plane. As shown in Figure 2.1, the lateral dynamic behaviour of the robot is therefore approximated by a rigid rod OD that is fixed on the ground at point O and is free to rotate about a horizontal longitudinal axis through O by an angle θ, and a uniform rigid rod AB that is attached to the first rod at point C_2 and is free to rotate with respect to the first rod by an angle ϕ about an axis through C_2 that is parallel to the axis of rotation of the first rod, under the influence of gravity. The centre of mass of rod OD is at point C_1 at distance h_1 from O and the centre of mass of rod AB is at point C_2 at distance h_2 from O. The mass and spin inertia (about its centre of mass) of rod OD are m_1 and I_1 respectively, and the corresponding quantities for rod AB are m_2 and I_2 respectively. A control torque T_c acting between the two rods at point C_2 provides the control action for balancing the robot.

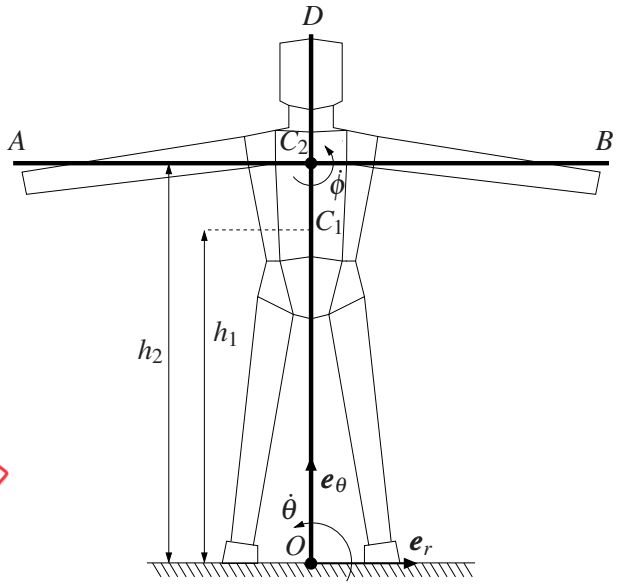


Figure 2.1 Humanoid robot abstraction by two rigid rods.

A moving Cartesian coordinate system with unit vectors $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ (see Figure 2.1) is used to analyse the motion of the two rods. The moving coordinate system is attached to the rod OD, it has its origin at O and it rotates by an angle θ. In the configuration shown in Figure 2.1, in which OD is vertical and AB is horizontal, $\theta = 0$ and $\phi = 0$.

a) Write the position and velocity vectors of the centre of mass of each of the two rods. [4]

b) Compute the total kinetic energy of the system. [5]

c) Calculate the total potential energy of the system, and hence determine the Lagrangian function. [4]

d) Use the Lagrangian approach to derive the equations of motion of the system in terms of the generalised coordinates. [7]

3. As in Q1, a compound pendulum of mass m consists of a symmetric rigid body (shown in Figure 3.1) and a horizontal massless plate (not shown in Figure 3.1). The plate is attached to point O in Figure 3.1 and is free to rotate by an angle ψ about the vertical Earth-fixed axis through O. The rigid body is fixed at one of its end points along its axis of symmetry (point O in Figure 3.1) to the plate and is free to rotate by an angle θ about a horizontal plate-fixed axis through O, under the influence of gravity. The centre of mass of the rigid body is on the axis of symmetry at distance f from point O.



Figure 3.1 Compound 2-degree-of-freedom pendulum.

A body-fixed Cartesian coordinate system has its origin at O, and its z-axis (Z') along the axis of symmetry of the body and its y-axis (Y') along the horizontal plate-fixed axis about which the rigid body rotates by the angle θ. The body-fixed x-, y-, and z-axes ($X'Y'Z'$) correspond to principal axes with principal moments of inertia I_{xx}, I_{yy} ($= I_{xx}$), and I_{zz} respectively. In the nominal configuration the body-fixed x-, y-, and z-axes are respectively aligned with the x-, y-, and z-axes (XYZ) of an Earth-fixed Cartesian coordinate system, in which the x-axis is horizontal and the z-axis points downwards. The angular configuration of the body is therefore given by the Euler angle convention, first rotation from its nominal configuration by angle ψ followed by angle θ rotation.

a) By using the standard single-axis-rotation transformation matrices calculate

i) the transformation matrix $\boldsymbol{A_1}$ from Earth-fixed to plate-fixed coordinates and the transformation matrix $\boldsymbol{A_2}$ from body-fixed to plate-fixed coordinates, and [3]

ii) the body angular momentum vector, $\boldsymbol{H}$, in the plate-fixed coordinate system. [5]

b) Compute in the plate-fixed coordinate system the total external moment $\boldsymbol{N}$ acting on the rigid body due to gravity and also in terms of a moment of magnitude M that maintains the rotational constraint of the compound pendulum. (Hint: the compound pendulum has 2 and not 3 rotational degrees of freedom.) [4]

c) By considering the rotational motion about the fixed point O, use the vectorial approach in plate-fixed coordinates to derive

i) the equations of motion of the compound pendulum, and [4]

ii) the moment M, in terms of the generalised coordinates [4]

4. As in Q2, it is desired to achieve lateral balancing of a humanoid robot by antisymmetric rotation of the robot arms in a vertical plane. As shown in Figure 4.1, the lateral dynamic behaviour of the robot is therefore approximated by a rigid rod OD that is fixed on the ground at point O and is free to rotate about a horizontal longitudinal axis through O by an angle θ, and a uniform rigid rod AB that is attached to the first rod at point C_2 and is free to rotate with respect to the first rod by an angle ϕ about an axis through C_2 that is parallel to the axis of rotation of the first rod, under the influence of gravity. The centre of mass of rod OD is at point C_1 at distance h_1 from O and the centre of mass of rod AB is at point C_2 at distance h_2 from O. The mass and spin inertia (about its centre of mass) of rod OD are m_1 and I_1 respectively, and the corresponding quantities for rod AB are m_2 and I_2 respectively. A control torque T_c acting between the two rods at point C_2 provides the control action for balancing the robot.

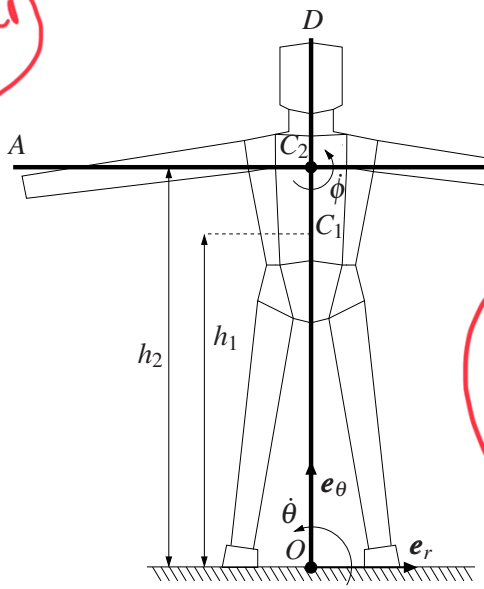


Figure 4.1 Humanoid robot abstraction by two rigid rods.

A moving Cartesian coordinate system with unit vectors $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ (see Figure 4.1) is used to analyse the motion of the two rods. The moving coordinate system is attached to the rod OD, it has its origin at O and it rotates by an angle θ. In the configuration shown in Figure 4.1, in which OD is vertical and AB is horizontal, $\theta = 0$ and $\phi = 0$.

a) Determine the generalised coordinates of the system. [2]

b) Write the acceleration vectors of the centre of mass of each of the rods. [3]

c) F_{r1} and $F_{\theta 1}$ are the unknown constraint forces on rod OD from Earth in the $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ directions respectively. F_{r2} and $F_{\theta 2}$ are the unknown constraint forces on rod AB from rod OD in the $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ directions respectively. Write the total force that acts on each of the rods in vector form. [3]

d) Use the vectorial approach to derive F_{r1}, $F_{\theta 1}$, F_{r2}, and $F_{\theta 2}$ in terms of the generalised coordinates. [6]

e) Use the vectorial approach to further derive the equations of motion of the system with respect to the generalised coordinates. [6]